##SQL CTE, JOINs & Subqueries (Hospital Management System)

```
Create database hospital_db;

use hospital_db;

CREATE TABLE Departments (
    department_id VARCHAR(10) PRIMARY KEY,
    department_name VARCHAR(50)
);

CREATE TABLE Doctors (
    doctor_id VARCHAR(10) PRIMARY KEY,
    doctor_name VARCHAR(50),
    department_id VARCHAR(10),
    specialization VARCHAR(50),
    FOREIGN KEY (department_id) REFERENCES
Departments(department_id)
);

CREATE TABLE Patients (
    patient_id VARCHAR(10) PRIMARY KEY,
    patient_name VARCHAR(50),
    gender VARCHAR(20),
    city VARCHAR(50),
    age INT
);

CREATE TABLE Appointments (
```

```sql
    appointment_id VARCHAR(10) PRIMARY KEY,
    patient_id VARCHAR(10),
    doctor_id VARCHAR(10),
    appointment_date DATE,
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)
);

CREATE TABLE Treatments (
    treatment_id VARCHAR(10) PRIMARY KEY,
    appointment_id VARCHAR(10),
    treatment_name VARCHAR(50),
    cost DECIMAL(15,2),
    FOREIGN KEY (appointment_id) REFERENCES
Appointments(appointment_id)
);

CREATE TABLE Medicines (
    medicine_id VARCHAR(10) PRIMARY KEY,
    medicine_name VARCHAR(50),
    price DECIMAL(15,2)
);

CREATE TABLE Prescriptions (
    prescription_id VARCHAR(10) PRIMARY KEY,
    appointment_id VARCHAR(10),
    medicine_id VARCHAR(10),
    quantity INT,
```

```
    FOREIGN KEY (appointment_id) REFERENCES
Appointments(appointment_id),
    FOREIGN KEY (medicine_id) REFERENCES Medicines(medicine_id)
);

CREATE TABLE Billing (
    bill_id VARCHAR(10) PRIMARY KEY,
    appointment_id VARCHAR(10),
    total_amount DECIMAL(15,2),
    payment_status VARCHAR(20),
    FOREIGN KEY (appointment_id) REFERENCES
Appointments(appointment_id)
);


INSERT INTO Departments (department_id, department_name) VALUES
('DEP01','Cardiology'),
('DEP02','Neurology'),
('DEP03','Orthopedics'),
('DEP04','General Medicine'),
('DEP05','Pediatrics');

INSERT INTO Doctors (doctor_id,doctor_name,department_id,specialization)
VALUES
('DOC001','Dr. Arun','DEP01','Cardiologist'),
('DOC002','Dr. Priya','DEP02','Neurologist'),
('DOC003','Dr. David','DEP03','Orthopedic'),
('DOC004','Dr. Meena','DEP04','General Physician'),
('DOC005','Dr. Ahmed','DEP05','Pediatrician');
```

```sql
INSERT INTO Patients (patient_id,patient_name,gender,city,age) VALUES
('PAT001','John','Male','Chennai',35),
('PAT002','Sarah','Female','Hyderabad',28),
('PAT003','David','Male','Mumbai',45),
('PAT004','Fatima','Female','Chennai',30),
('PAT005','Rahul','Male','Bangalore',40),
('PAT006','Anita','Female','Delhi',32),
('PAT007','Kumar','Male','Coimbatore',29),
('PAT008','Priya','Female','Salem',26),
('PAT009','Ravi','Male','Madurai',38),
('PAT010','Ayesha','Female','Kochi',34);


INSERT INTO Appointments
(appointment_id,patient_id,doctor_id,appointment_date)
VALUES
('APP001','PAT001','DOC001','2025-01-10'),
('APP002','PAT002','DOC002','2025-01-11'),
('APP003','PAT003','DOC003','2025-01-12'),
('APP004','PAT004','DOC004','2025-01-13'),
('APP005','PAT005','DOC001','2025-01-15'),
('APP006','PAT006','DOC005','2025-01-18'),
('APP007','PAT007','DOC004','2025-01-20'),
('APP008','PAT008','DOC003','2025-01-22');


INSERT INTO Treatments
(treatment_id,appointment_id,treatment_name,cost) VALUES
('TR001','APP001','ECG',2500),
```

```sql
('TR002','APP002','Brain Scan',8000),
('TR003','APP003','Bone Surgery',55000),
('TR004','APP004','General Checkup',1200),
('TR005','APP005','Heart Scan',6000),
('TR006','APP006','Vaccination',1500),
('TR007','APP007','Blood Test',900),
('TR008','APP008','X-Ray',1800);


INSERT INTO Medicines (medicine_id,medicine_name,price) VALUES
('MED001','Paracetamol',50),
('MED002','Amoxicillin',120),
('MED003','Vitamin C',80),
('MED004','Ibuprofen',100),
('MED005','Insulin',500);

INSERT INTO Prescriptions
(prescription_id,appointment_id,medicine_id,quantity) VALUES
('PRE001','APP001','MED001',10),
('PRE002','APP002','MED002',7),
('PRE003','APP003','MED004',15),
('PRE004','APP004','MED003',12),
('PRE005','APP005','MED005',5),
('PRE006','APP006','MED001',8),
('PRE007','APP007','MED003',6),
('PRE008','APP008','MED004',9);
```

```sql
INSERT INTO Billing (bill_id,appointment_id,total_amount,payment_status)
VALUES
('B001','APP001',3000,'Paid'),
('B002','APP002',9000,'Paid'),
('B003','APP003',60000,'Pending'),
('B004','APP004',1500,'Paid'),
('B005','APP005',7000,'Pending'),
('B006','APP006',1800,'Paid'),
('B007','APP007',1200,'Paid'),
('B008','APP008',2200,'Pending');
```

```sql
select * from Departments
```

Result Grid | Filter Rows:

department_id	department_name
DEP01	Cardiology
DEP02	Neurology
DEP03	Orthopedics
DEP04	General Medicine
DEP05	Pediatrics
NULL	NULL

```sql
select * from Doctors
```

Result Grid | Filter Rows: | Edit:

doctor_id	doctor_name	department_id	specialization
DOC001	Dr. Arun	DEP01	Cardiologist
DOC002	Dr. Priya	DEP02	Neurologist
DOC003	Dr. David	DEP03	Orthopedic
DOC004	Dr. Meena	DEP04	General Physician
DOC005	Dr. Ahmed	DEP05	Pediatrician
NULL	NULL	NULL	NULL

DEP04

Doctors 53

select * from Patients

Result Grid | Filter Rows: | Edit:

patient_id	patient_name	gender	city	age
PAT001	John	Male	Chennai	35
PAT002	Sarah	Female	Hyderabad	28
PAT003	David	Male	Mumbai	45
PAT004	Fatima	Female	Chennai	30
PAT005	Rahul	Male	Bangalore	40
PAT006	Anita	Female	Delhi	32
PAT007	Kumar	Male	Coimbatore	29
PAT008	Priya	Female	Salem	26
PAT009	Ravi	Male	Madurai	38
PAT010	Ayesha	Female	Kochi	34
NULL	NULL	NULL	NULL	NULL

Patients 54 ×

select * from Appointments

Result Grid | Filter Rows: | Edit:

appointment_id	patient_id	doctor_id	appointment_date
APP001	PAT001	DOC001	2025-01-10
APP002	PAT002	DOC002	2025-01-11
APP003	PAT003	DOC002	2025-01-12
APP004	PAT004	DOC004	2025-01-13
APP005	PAT005	DOC001	2025-01-15
APP006	PAT006	DOC005	2025-01-18
APP007	PAT007	DOC004	2025-01-20
APP008	PAT008	DOC003	2025-01-22
NULL	NULL	NULL	NULL

Appointments 55 ×

select * from Treatments

Result Grid | Filter Rows: | Edit:

treatment_id	appointment_id	treatment_name	cost
TR001	APP001	ECG	2500.00
TR002	APP002	Brain Scan	8000.00
TR003	APP003	Bone Surgery	55000.00
TR004	APP004	General Checkup	1200.00
TR005	APP005	Heart Scan	6000.00
TR006	APP006	Vaccination	1500.00
TR007	APP007	Bloo Vaccination	900.00
TR008	APP008	X-Ray	1800.00
NULL	NULL	NULL	NULL

select * from Medicines

Result Grid | Filter Rows: | Edi

medicine_id	medicine_name	price
MED001	Paracetamol	50.00
MED002	Amoxicillin	120.00
MED003	Vitamin C	80.00
MED004	Ibuprofen	100.00
MED005	Insulin	500.00
NULL	NULL	NULL

select * from Prescriptions

Result Grid | Filter Rows: | Edit:

prescription_id	appointment_id	medicine_id	quantity
PRE001	APP001	APP001	10
PRE002	APP002	MED002	7
PRE003	APP003	MED004	15
PRE004	APP004	MED003	12
PRE005	APP005	MED005	5
PRE006	APP006	MED001	8
PRE007	APP007	MED003	6
PRE008	APP008	MED004	9
NULL	NULL	NULL	NULL

select * from Billing

bill_id	appointment_id	total_amount	payment_status
B001	APP001	3000.00	Paid
B002	APP002	9000.00	Paid
B003	APP003	60000.00	Pending
B004	APP004	1500.00	Paid
B005	APP005	7000.00	Pending
B006	APP006	1800.00	Paid
B007	APP007	1200.00	Paid
B008	APP008	2200.00	Pending
NULL	NULL	NULL	NULL

Display all patient details using a CTE

```
WITH Patient_CTE AS
(
    SELECT
        patient_id,
        patient_name,
        gender,
        city,
        age
    FROM Patients
)
SELECT *
FROM Patient_CTE;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	patient_id	patient_name	gender	city	age
▶	PAT001	John	Male	Chennai	35
	PAT002	Sarah	Female	Hyderabad	28
	PAT003	David	Male	Mumbai	[illegible]
	PAT004	Fatima	Female	Chennai	30
	PAT005	Rahul	Male	Bangalore	40
	PAT006	Anita	Female	Delhi	32
	PAT007	Kumar	Male	Coimbatore	29
	PAT008	Priya	Female	Salem	26
	PAT009	Ravi	Male	Madurai	38
	PAT010	Ayesha	Female	Kochi	34

Display patients above the average age using a CTE

```
WITH AvgAge_CTE AS
(
    SELECT AVG(age) AS AverageAge
    FROM Patients
)
SELECT
    p.patient_id,
    p.patient_name,
    p.gender,
    p.city,
    p.age
FROM Patients p
CROSS JOIN AvgAge_CTE a
WHERE p.age > a.AverageAge;
```

OUTPUT

patient_id	patient_name	gender	city	age
PAT001	John	Male	Chennai	35
PAT003	David	Male	Mumbai	45
PAT005	Rahul	Male	Bangalore	40
PAT009	Ravi	Male	Madurai	38
PAT010	Ayesha	Female	Kochi	34

##Find doctors who handled more than one appointment using a CTE

```
WITH DoctorAppointments AS
(
    SELECT doctor_id,
           COUNT(*) AS TotalAppointments
    FROM Appointments
    GROUP BY doctor_id
)
SELECT d.doctor_name,
       da.TotalAppointments
FROM DoctorAppointments da
JOIN Doctors d
ON da.doctor_id = d.doctor_id
WHERE da.TotalAppointments > 1;
```

OUTPUT

doctor_name	TotalAppointments
Dr. Arun	2
Dr. David	2
Dr. Meena	2

Display the department-wise appointment count using a CTE

```
WITH DeptAppointments AS
(
   SELECT dep.department_name,
         COUNT(a.appointment_id) AS AppointmentCount
   FROM Departments dep
   JOIN Doctors d
      ON dep.department_id = d.department_id
   JOIN Appointments a
      ON d.doctor_id = a.doctor_id
   GROUP BY dep.department_name
)
SELECT *
FROM DeptAppointments;
```

OUTPUT

department_name	AppointmentCount
Cardiology	2
Neurology	1
Orthopedics	2
General Medicine	2
Pediatrics	1

Find the top three treatment costs using a CTE

```
WITH TreatmentRank AS
(
    SELECT treatment_name,
           cost,
           DENSE_RANK() OVER(ORDER BY cost DESC) AS RankNo
    FROM Treatments
)
SELECT *
FROM TreatmentRank
WHERE RankNo <= 3;
```

OUTPUT

treatment_name	cost	RankNo
Bone Surgery	55000.00	1
Brain Scan	8000.00	2
Heart Scan	6000.00	3

Display patients who paid more than the average bill

```
WITH AvgBill AS
(
    SELECT AVG(total_amount) AS AvgBill
    FROM Billing
)
SELECT p.patient_name,
       b.total_amount
FROM Patients p
JOIN Appointments a
ON p.patient_id = a.patient_id
JOIN Billing b
ON a.appointment_id = b.appointment_id
JOIN AvgBill ab
ON b.total_amount > ab.AvgBill;
```

OUTPUT

Result Grid | Filter Rows:

patient_name	total_amount
David	60000.00

Display total revenue generated by each department

```
WITH DepartmentRevenue AS
(
    SELECT dep.department_name,
           SUM(b.total_amount) AS Revenue
    FROM Departments dep
    JOIN Doctors d
        ON dep.department_id = d.department_id
    JOIN Appointments a
        ON d.doctor_id = a.doctor_id
    JOIN Billing b
        ON a.appointment_id = b.appointment_id
    GROUP BY dep.department_name
)
SELECT *
FROM DepartmentRevenue;
```

OUTPUT

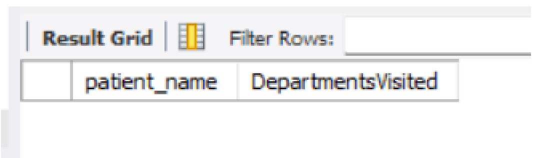
Result Grid | Filter Rows:

department_name	Revenue
Cardiology	10000.00
Neurology	9000.00
Orthopedics	62200.00
General Medicine	2700.00
Pediatrics	1800.00

Find patients who visited multiple departments

```
WITH PatientDept AS
(
   SELECT p.patient_name,
          COUNT(DISTINCT d.department_id) AS DepartmentsVisited
   FROM Patients p
   JOIN Appointments a
      ON p.patient_id = a.patient_id
   JOIN Doctors d
      ON a.doctor_id = d.doctor_id
   GROUP BY p.patient_name
)
SELECT *
FROM PatientDept
WHERE DepartmentsVisited > 1;
```

OUTPUT



Display doctor-wise treatment counts

```
WITH DoctorTreatment AS
(
   SELECT d.doctor_name,
          COUNT(t.treatment_id) AS TotalTreatments
   FROM Doctors d
```

```
    JOIN Appointments a
        ON d.doctor_id = a.doctor_id
    JOIN Treatments t
        ON a.appointment_id = t.appointment_id
    GROUP BY d.doctor_name
)
SELECT *
FROM DoctorTreatment;
```

OUTPUT

doctor_name	TotalTreatments
Dr. Arun	2
Dr. Priya	1
Dr. David	2
Dr. Meena	2
Dr. Ahmed	1

Find patients with pending bills

```
WITH PendingBills AS
(
    SELECT p.patient_name,
           b.total_amount,
           b.payment_status
    FROM Patients p
    JOIN Appointments a
        ON p.patient_id = a.patient_id
    JOIN Billing b
        ON a.appointment_id = b.appointment_id
    WHERE b.payment_status = 'Pending'
```

)

SELECT *

FROM PendingBills;

OUTPUT

patient_name	total_amount	payment_status
David	60000.00	Pending
Rahul	7000.00	Pending
Priya	2200.00	Pending

Display patient name, doctor name, and department

```
SELECT
    p.patient_name,
    d.doctor_name,
    dep.department_name
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Doctors d
    ON a.doctor_id = d.doctor_id
INNER JOIN Departments dep
    ON d.department_id = dep.department_id;
```

OUTPUT

patient_name	doctor_name	department_name
John	Dr. Arun	Cardiology
Rahul	Dr. Arun	Cardiology
Sarah	Dr. Priya	Neurology
David	Dr. David	Orthopedics
Priya	Dr. David	Orthopedics
Fatima	Dr. Meena	General Medicine
Kumar	Dr. Meena	General Medicine
Anita	Dr. Ahmed	Pediatrics

Display patient name with treatment details

```
SELECT
    p.patient_name,
    t.treatment_name,
    t.cost
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Treatments t
    ON a.appointment_id = t.appointment_id;
```

OUTPUT

Result Grid | Filter Rows:

patient_name	treatment_name	cost
John	ECG	2500.00
Sarah	Brain Scan	8000.00
David	Bone Surgery	55000.00
Fatima	General Checkup	1200.00
Rahul	Heart Scan	6000.00
Anita	Vaccination	1500.00
Kumar	Blood Test	900.00
Priya	X-Ray	1800.00

Display patient name with medicine details

```
SELECT
    p.patient_name,
    m.medicine_name,
    m.price,
    pr.quantity
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Prescriptions pr
    ON a.appointment_id = pr.appointment_id
INNER JOIN Medicines m
    ON pr.medicine_id = m.medicine_id;
```

OUTPUT

patient_name	medicine_name	price	quantity
John	Paracetamol	50.00	10
Anita	Paracetamol	50.00	8
Sarah	Amoxicillin	120.00	7
Fatima	Vitamin C	80.00	12
Kumar	Vitamin C	80.00	6
David	Ibuprofen	100.00	15
Priya	Ibuprofen	100.00	9
Rahul	Insulin	500.00	5

Display department-wise total billing

```
SELECT
    dep.department_name,
    SUM(b.total_amount) AS Total_Billing
FROM Departments dep
INNER JOIN Doctors d
    ON dep.department_id = d.department_id
INNER JOIN Appointments a
    ON d.doctor_id = a.doctor_id
INNER JOIN Billing b
    ON a.appointment_id = b.appointment_id
GROUP BY dep.department_name;
```

OUTPUT

department_name	Total_Billing
Cardiology	10000.00
Neurology	9000.00
Orthopedics	62200.00
General Medicine	2700.00
Pediatrics	1800.00

Find the highest billed patient

```
SELECT
    p.patient_name,
    b.total_amount
FROM Patients p
JOIN Appointments a
    ON p.patient_id = a.patient_id
JOIN Billing b
    ON a.appointment_id = b.appointment_id
ORDER BY b.total_amount DESC
LIMIT 1;
```

OUTPUT

patient_name	total_amount
David	60000.00

Display doctor-wise revenue

```
SELECT
    d.doctor_name,
    SUM(b.total_amount) AS Revenue
FROM Doctors d
INNER JOIN Appointments a
    ON d.doctor_id = a.doctor_id
INNER JOIN Billing b
    ON a.appointment_id = b.appointment_id
GROUP BY d.doctor_name
ORDER BY Revenue DESC;
```

OUTPUT

Result Grid | Filter Rows:

doctor_name	Revenue
Dr. David	62200.00
Dr. Arun	10000.00
Dr. Priya	9000.00
Dr. Meena	2700.00
Dr. Ahmed	1800.00

Display all appointments with billing status

```
SELECT
    a.appointment_id,
    p.patient_name,
    d.doctor_name,
    a.appointment_date,
    b.total_amount,
    b.payment_status
```

```
FROM Appointments a
INNER JOIN Patients p
    ON a.patient_id = p.patient_id
INNER JOIN Doctors d
    ON a.doctor_id = d.doctor_id
LEFT JOIN Billing b
    ON a.appointment_id = b.appointment_id
ORDER BY a.appointment_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

appointment_id	patient_name	doctor_name	appointment_date	total_amount	payment_status
APP001	John	Dr. Arun	2025-01-10	3000.00	Paid
APP002	Sarah	Dr. Priya	2025-01-11	9000.00	Paid
APP003	David	Dr. David	2025-01-12	60000.00	Pending
APP004	Fatima	Dr. Meena	2025-01-13	1500.00	Paid
APP005	Rahul	Dr. Arun	2025-01-15	7000.00	Pending
APP006	Anita	Dr. Ahmed	2025-01-18	1800.00	Paid
APP007	Kumar	Dr. Meena	2025-01-20	1200.00	Paid
APP008	Priya	Dr. David	2025-01-22	2200.00	Pending

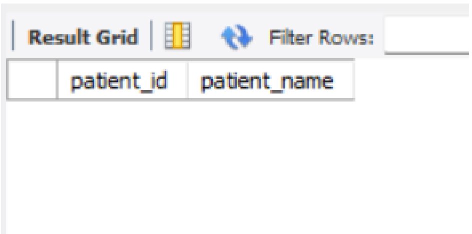
Display patients without prescriptions

```
SELECT
    p.patient_id,
    p.patient_name
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
LEFT JOIN Prescriptions pr
    ON a.appointment_id = pr.appointment_id
WHERE pr.prescription_id IS NULL;
```

Display doctors without appointments

```
SELECT
    d.doctor_id,
    d.doctor_name,
    d.specialization
FROM Doctors d
LEFT JOIN Appointments a
    ON d.doctor_id = a.doctor_id
WHERE a.appointment_id IS NULL;
```

OUTPUT



Display department-wise patient count

```
SELECT
    dep.department_name,
    COUNT(DISTINCT a.patient_id) AS Total_Patients
FROM Departments dep
INNER JOIN Doctors d
    ON dep.department_id = d.department_id
INNER JOIN Appointments a
    ON d.doctor_id = a.doctor_id
GROUP BY dep.department_name
ORDER BY Total_Patients DESC;
```

OUTPUT

doctor_id	doctor_name	specialization

Find patients whose bill is greater than the average bill

```
SELECT
    p.patient_name,
    b.total_amount
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Billing b
    ON a.appointment_id = b.appointment_id
WHERE b.total_amount >
(
    SELECT AVG(total_amount)
    FROM Billing
);
```

OUTPUT

department_name	Total_Patients
Cardiology	2
General Medicine	2
Orthopedics	2
Neurology	1
Pediatrics	1

Display doctors earning the highest department revenue

```
WITH DoctorRevenue AS
(
    SELECT
        dep.department_name,
        d.doctor_name,
        SUM(b.total_amount) AS Revenue,
        DENSE_RANK() OVER
        (PARTITION BY dep.department_name ORDER BY 
SUM(b.total_amount) DESC) AS RankNo
    FROM Departments dep
    INNER JOIN Doctors d
        ON dep.department_id = d.department_id
    INNER JOIN Appointments a
        ON d.doctor_id = a.doctor_id
    INNER JOIN Billing b
        ON a.appointment_id = b.appointment_id
    GROUP BY dep.department_name,
             d.doctor_name)
SELECT
    department_name,
    doctor_name,
    Revenue
FROM DoctorRevenue
WHERE RankNo = 1;
```

OUTPUT

Result Grid | Filter Rows:

patient_name	total_amount
David	60000.00

Find patients prescribed the most expensive medicine

```
SELECT
    p.patient_name,
    m.medicine_name,
    m.price
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Prescriptions pr
    ON a.appointment_id = pr.appointment_id
INNER JOIN Medicines m
    ON pr.medicine_id = m.medicine_id
WHERE m.price =
(
    SELECT MAX(price)
    FROM Medicines
);
```

OUTPUT

Display departments having above-average revenue

```
WITH DepartmentRevenue AS
(
    SELECT
        dep.department_name,
        SUM(b.total_amount) AS Revenue
    FROM Departments dep
    INNER JOIN Doctors d
        ON dep.department_id = d.department_id
    INNER JOIN Appointments a
        ON d.doctor_id = a.doctor_id
    INNER JOIN Billing b
        ON a.appointment_id = b.appointment_id
    GROUP BY dep.department_name
)

SELECT *
FROM DepartmentRevenue
WHERE Revenue >
(
    SELECT AVG(Revenue)
    FROM DepartmentRevenue
);
```

OUTPUT

department_name	doctor_name	Revenue
Cardiology	Dr. Arun	10000.00
General Medicine	Dr. Meena	2700.00
Neurology	Dr. Priya	9000.00
Orthopedics	Dr. David	62200.00
Pediatrics	Dr. Ahmed	1800.00

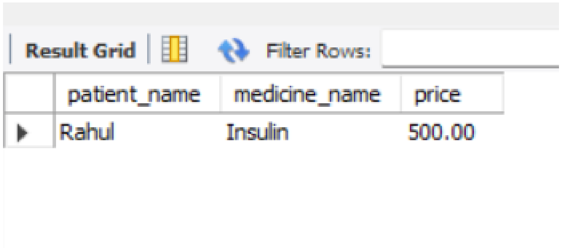
Find doctors who handled the maximum appointments

```
WITH DoctorAppointments AS
(
    SELECT
        d.doctor_name,
        COUNT(a.appointment_id) AS TotalAppointments,
        DENSE_RANK() OVER
        (
            ORDER BY COUNT(a.appointment_id) DESC
        ) AS RankNo
    FROM Doctors d
    LEFT JOIN Appointments a
        ON d.doctor_id = a.doctor_id
    GROUP BY d.doctor_name
)

SELECT
    doctor_name,
    TotalAppointments
```

```sql
FROM DoctorAppointments
WHERE RankNo = 1;
```

OUTPUT



patient_name	medicine_name	price
Rahul	Insulin	500.00

Display patients with treatment costs greater than the average treatment cost

```sql
SELECT
    p.patient_name,
    t.treatment_name,
    t.cost
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Treatments t
    ON a.appointment_id = t.appointment_id
WHERE t.cost >
(
    SELECT AVG(cost)
    FROM Treatments
);
```

OUTPUT

Result Grid | Filter Rows:

department_name	Revenue
Orthopedics	62200.00

Find medicines prescribed more than the average quantity

```
SELECT
    m.medicine_name,
    pr.quantity
FROM Medicines m
INNER JOIN Prescriptions pr
    ON m.medicine_id = pr.medicine_id
WHERE pr.quantity >
(
    SELECT AVG(quantity)
    FROM Prescriptions
);
```

OUTPUT

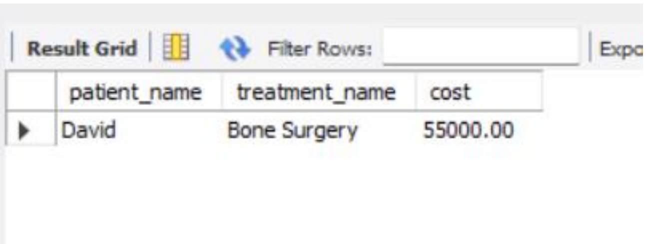
Result Grid | Filter Rows:

doctor_name	TotalAppointments
Dr. Arun	2
Dr. David	2
Dr. Meena	2

Display patient-wise total hospital expenditure

```
SELECT
    p.patient_name,
    SUM(b.total_amount) AS TotalExpenditure
FROM Patients p
INNER JOIN Appointments a
    ON p.patient_id = a.patient_id
INNER JOIN Billing b
    ON a.appointment_id = b.appointment_id
GROUP BY p.patient_name
ORDER BY TotalExpenditure DESC;
```

OUTPUT



patient_name	treatment_name	cost
David	Bone Surgery	55000.00

Find departments with no pending payments

```
SELECT
    dep.department_name
FROM Departments dep
INNER JOIN Doctors d
    ON dep.department_id = d.department_id
INNER JOIN Appointments a
    ON d.doctor_id = a.doctor_id
INNER JOIN Billing b
```

```
    ON a.appointment_id = b.appointment_id
GROUP BY dep.department_name
HAVING SUM(
    CASE
        WHEN b.payment_status = 'Pending' THEN 1
        ELSE 0
    END
) = 0;
```

OUTPUT

medicine_name	quantity
Paracetamol	10
Ibuprofen	15
Vitamin C	12

Generate a complete hospital report using CTEs, JOINs, and Subqueries

```
WITH HospitalReport AS
(
    SELECT
        p.patient_name,
        d.doctor_name,
        dep.department_name,
        t.treatment_name,
        t.cost,
        m.medicine_name,
        pr.quantity,
```

```
        b.total_amount,
        b.payment_status
    FROM Patients p
    INNER JOIN Appointments a
        ON p.patient_id = a.patient_id
    INNER JOIN Doctors d
        ON a.doctor_id = d.doctor_id
    INNER JOIN Departments dep
        ON d.department_id = dep.department_id
    LEFT JOIN Treatments t
        ON a.appointment_id = t.appointment_id
    LEFT JOIN Prescriptions pr
        ON a.appointment_id = pr.appointment_id
    LEFT JOIN Medicines m
        ON pr.medicine_id = m.medicine_id
    LEFT JOIN Billing b
        ON a.appointment_id = b.appointment_id
)

SELECT *
FROM HospitalReport
ORDER BY total_amount DESC;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_name	doctor_name	department_name	treatment_name	cost	medicine_name	quantity	total_amount	payment_status
David	Dr. David	Orthopedics	Bone Surgery	55000.00	Ibuprofen	15	60000.00	Pending
Sarah	Dr. Priya	Neurology	Brain Scan	8000.00	Amoxicillin	7	9000.00	Paid
Rahul	Dr. Arun	Cardiology	Heart Scan	6000.00	Insulin	5	7000.00	Pending
John	Dr. Arun	Cardiology	ECG	2500.00	Paracetamol	10	3000.00	Paid
Priya	Dr. David	Orthopedics	X-Ray	1800.00	Ibuprofen	9	2200.00	Pending
Anita	Dr. Ahmed	Pediatrics	Vaccination	1500.00	Paracetamol	8	1800.00	Paid
Fatima	Dr. Meena	General Medicine	General Checkup	1200.00	Vitamin C	12	1500.00	Paid
Kumar	Dr. Meena	General Medicine	Blood Test	900.00	Vitamin C	6	1200.00	Paid